



Effect of noise floor in room impulse response on speech perception under spherical harmonics-based spatial sound reproduction

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Abstract

The current study investigates the effect of noise floor in measured room impulse responses (RIR) on the reproducibility of speech perception under spherical harmonics-based spatial sound reproduction. Subjective listening test measuring the intelligibility of speech in noise was conducted under the spatial sound reproduction implemented using practically measured RIR with varying level of noise floor. The same test was also conducted in the real rooms where the RIR were measured. The comparison of the experimental results from the spatial sound reproduction and the real room suggests using measured RIR with low noise floor contributes to reproducing speech perception in real rooms accurately when the room is highly reverberant. It also has an effect to improve the reproducibility when the sound sources are located at 5 m but not at 2 m. Truncating RIR to further remove the noise floor mostly did not help improve the reproducibility regardless of the acoustics of the room.

Index Terms: spatial sound reproduction, spherical harmonics, speech intelligibility, room impulse response, noise floor, reproducibility

1. Introduction

Spatial sound reproduction virtually replicates the auditory perception of sound in an arbitrary acoustic environment such as sound source location and spatial extent of the space [1]. As more equipment and tools become available to implement spatial sound reproduction, its application areas have also been extending rapidly. Apart from various commercial applications, thanks to its high reproducibility of auditory perception, there have been growing number of reports in hearing research that use spatial sound reproduction to examine humans' auditory perception in varying acoustic environment [2, 3, 4]. This emerging research methodology enables studying how humans perceive speech in practical acoustic environments in a controlled manner.

Many of such hearing studies using spatial sound reproduction adopted spherical harmonics-based (also known as Ambisonics) techniques [2, 5, 6, 7, 8, 9]. When spherical harmonics-based spatial sound reproduction is implemented, the acoustical properties of the space are often specified by using room impulse responses (RIR), which encapsulates the information about the propagation of sound in the space. While RIR can be either numerically simulated based on the physical model of the space or practically measured in the real space [6], Ahrens *et al.* [10] found that speech perception in the real space was replicated most accurately when practically measured RIR were utilised. However, in practical spaces where surrounding environment is normally uncontrollable, quality of RIR measured can be often compromised due to various ambient noise

in the space. RIR measured in such a noisy environment typically suffers from non-decaying “noise floor” [11], which may produce infinite reverberation effect [12] in the stimuli rendered from the spatial sound reproduction. Using such low quality RIR for implementing spatial sound reproduction would eventually degrade the reproducibility of speech perception in the original space and invalidate/limit the value of the experimental results collected under spatial sound reproduction. While some previous studies investigated the reproducibility of speech perception using spatial sound reproduction [10, 13, 14], there has been no report of studies examining the effect of the noise floor in RIR on the reproducibility of speech perception.

Various techniques have been studied for minimising the noise floor in measured RIR. An approach to address the noise floor is using measurement methods known to be robust to specific type of ambient noise [15, 16]. Another approach is post-processing the measured RIR, which removes the noise floor from the measurements [17], such as truncating the RIR to remove the parts impacted by the noise floor [18]. However, most of these techniques have been developed for improving the accuracy of estimating acoustical parameters, e.g. reverberation time, which are typically calculated from measured RIR. The effect on speech perception when the processed RIR are used for spatial sound reproduction is yet unknown.

The current study investigates the effect of noise floor in measured RIR on the reproducibility of speech perception under spherical harmonics-based spatial sound reproduction. The study particularly aims to answer the questions: i) how noise floor in measured RIR used for spatial sound reproduction affects the reproducibility of speech perception in the original space, and ii) if removing noise floor by truncating RIR would improve the reproducibility of speech perception. The study intends to shed light on this niche but non-trivial challenge in using spatial sound reproduction for hearing research.

2. Methodology

Subjective listening tests were conducted to evaluate the speech intelligibility under spherical harmonics-based spatial sound reproduction (called *sound reproduction test*) using RIR with varying level of noise floor. To obtain baseline performance, the same listening tests were also performed in the actual rooms where the RIR were measured (called *Real room test*). To evaluate the performance in terms of replicating spatial extent of sound in the original rooms, the experimental design included the reverberation time in the room, distance and angles of the target speech from the listener, as varying parameters. This section provides the summary of spatial sound reproduction implemented including how the RIR with varying level of noise floor were generated, the design of the listening tests, and the

Table 1: *Acoustical properties of the rooms.*

	T_{20} (s)	L_{Aeq} (dB)	Volume (m ³)	C_{50} (dB)	
				2 m	5 m
Seminar room	0.7	31	~400	15.9	6.8
Chapel	1.8	37	~1200	10.3	3.3

statistical methods used for analysing the experimental results.

2.1. Spatial sound reproduction

For implementing spatial sound reproduction, the current study followed the set up used in the previous study [8] and employed the techniques available in the SPARTA plugin¹, which is an open-source package for implementing spherical harmonics-based sound reproduction. The system takes RIR measured by a spherical microphone array as input, which are encoded to the representation in the spherical harmonics domain by the Array2SH plugin [19]. These RIR represented by spherical harmonics are then decoded using the AmbiDEC plugin. The decoded RIR are convolved with arbitrary audio files to generate the stimuli (see Section 2.3) to be rendered from the 32-ch loudspeaker array installed in the anechoic chamber of the Acoustics Laboratory at the University of Auckland.

2.2. Room acoustics and impulse response measurement

The current study utilised two rooms, seminar room and chapel both located at the University of Auckland, for measuring the RIR to replicate their acoustics in the sound reproduction test as well as conducting the real room test. In each room, the distance between the positions of the speaker and listener was varied between 2 m and 5 m. Hence, overall speech intelligibility was measured in four acoustics (2 rooms $\times$ 2 distances) with distinct speech clarity (C_{50}) [1], which are known to correlate well with speech intelligibility in rooms. The acoustical properties of the rooms are summarised in Table 1.

The RIR were measured in each acoustic condition. A 32-ch spherical microphone array (MH Acoustics Eigenmike) was placed at the centre of each room and oriented at a fixed angle defined as 0°. A loudspeaker (Genelec 8020D) was placed at four different angles: -45°, 0°, 45° and 180°, either 2 or 5 m away from the microphone array. Figure 1 shows the positions of the loudspeaker and microphone array along with the dimension of the rooms. The RIR were measured at 48 kHz sampling rate using the open software tool ScanIR [20, 21] by playing a swept sine signal from the loudspeaker. The length of the measured RIR was 2 s for the seminar room and 2.5 s for the chapel.

Overall, three types of RIR were examined in this study. The two were RIR with different level of noise floor acquired by varying the repetition of the swept sine signal for synchronous averaging [16], a well-known technique to lower the noise floor, between 6 and 40 resulting in decreasing the level of noise floor by approximately 15 dB. These RIR with low and high level of noise floor are called *Low* (noise floor) and *High* (noise floor), respectively. In addition, the Low RIR truncated at the intersection time, the time instant where the noise floor exceeds the energy decay [18], was also examined. This RIR is called *Low truncated*. Figure 2 shows the energy decay curves [1] of the RIR measured at 0° in each acoustic condition encoded to the 0-th order spherical harmonics (i.e. omni-directional), with the positions of the intersection time.

¹<https://leomccormack.github.io/sparta-site/>

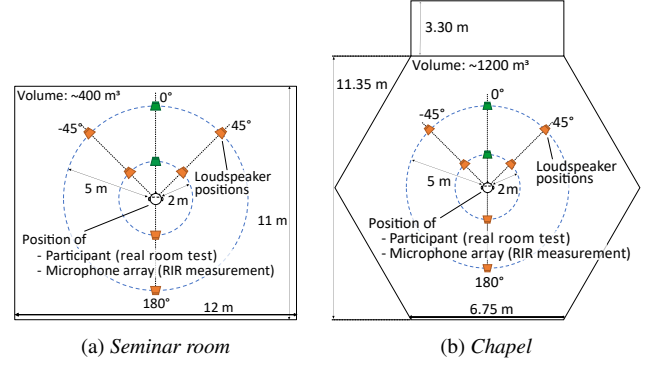


Figure 1: *Dimension and volume of the rooms used in the current study with the positions of sound sources and participants. The location of the target speech was fixed at one of the green loudspeaker icons, whereas the location of the noise source was varied between the locations of both green and orange loudspeaker icons. The distance of the target speech and noise was kept the same, i.e. either 2 m or 5 m.*

2.3. Stimuli

The test stimuli consisted of target speech and babble noise rendered simultaneously to simulate speech in noise. The target speech was selected from the Bamford-Kowal-Bench (BKB) sentence lists [22] in the SPANZ corpus [23], while the noise was sourced from the NOISEX-92 corpus [24], both resampled to 48 kHz. The sound sources were presented at the sound levels of 50(±1) dBA for speech and 53(±1) dBA for noise, both measured at the listener's position, resulting in a target-to-masker ratio (TMR) of -3 dB. In both real and sound reproduction tests, the position of the target speech was fixed at 0°, while that of the noise was varied across the four spatial angles specified in Figure 1. Four repetitions were made by varying the speech sentence for each condition, and the target speech and babble noise shared the same distance from the listener (2 m or 5 m).

2.4. Participants

The listening tests recruited in total 54 native New Zealand English listeners in Auckland, New Zealand, who are categorised into four groups. Participants in Group 1 ($N = 16$, mean age = 21.3, $sd = 0.77$) undertook the test using Low and Low truncated RIR whereas those in Group 2 ($N = 16$, mean age = 21.7, $sd = 1.66$) undertook the test using High RIR. Participants in Group 3 ($N = 11$, mean age = 25.5, $sd = 3.06$) undertook the test in the real seminar room and those in Group 4 ($N = 11$, mean age = 21.8, $sd = 2.86$) undertook the test in the real chapel. All participants self-reported no hearing impairment at the time of participating the experiment.

2.5. Procedure

In the listening test, participants were asked to transcribe speech sentences they listened to under noise through the graphical user interface displayed on a monitor located in 0°. While the speech sentences and their order were kept consistent for all participants, the combination of parameters (noise source angles and source distances) applied to each speech sentence was randomised between participants except the source distance of the real room test due to logistical difficulty. For the sound reproduction test, participants were seated at the centre of the

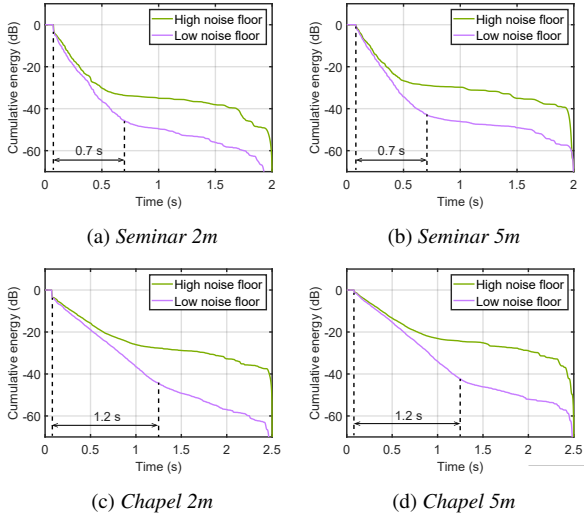


Figure 2: Energy decay curves of the room impulse responses (encoded to 0-th order spherical harmonics) measured at 0° . The points the RIR with low noise floor level were truncated are shown by the dashed line.

loudspeaker array, by adjusting the height of the seat to ensure their ears were inside the sweet spot of the spatial sound reproduction (i.e. centre of the array). They were instructed to hold this position during the test and keep their head facing towards the monitor. For the real room test, the participants were seated at the positions where the microphone array was placed for the RIR measurements discussed in Sec. 2.2. Loudspeakers were also placed at the same position used for the RIR measurements which rendered the speech and/or noise stimuli.

The participants typed in their responses through a keyboard while looking at the monitor in front of them (0°). Participants could take as long as they needed to type in their answer, however, they were not allowed to listen to the sentences more than once. They were encouraged to take a break whenever they needed to. The length of the listening tests ranged from 30 - 45 minutes for each participant. Participants were given a voucher to thank their participation to the experiment. The procedure was approved by the University of Auckland Human Ethics Committee (#024182).

2.6. Marking and statistical analysis

The participants' responses were marked according to the marking schedule used in the previous study using spatial sound reproduction for hearing research [8]. The resultant proportion correct scores normalised from 0 to 1, called *proportion correct* hereafter, were statistically analysed separately for each room by linear mixed effect models using the *lme4* package in R [25] with the noise source angle, source distance and environment (virtual sound environments realised by three types of RIR plus the real room) as fixed effects and participant ID as a random effect. The model fitting was conducted by the step function from *lmerTest* [26], by including interactions between multiple factors when the fitness of the model was improved. Post-hoc analysis for pairwise comparison was implemented by *emmeans* package [27] with correction of p -values by Tukey method.

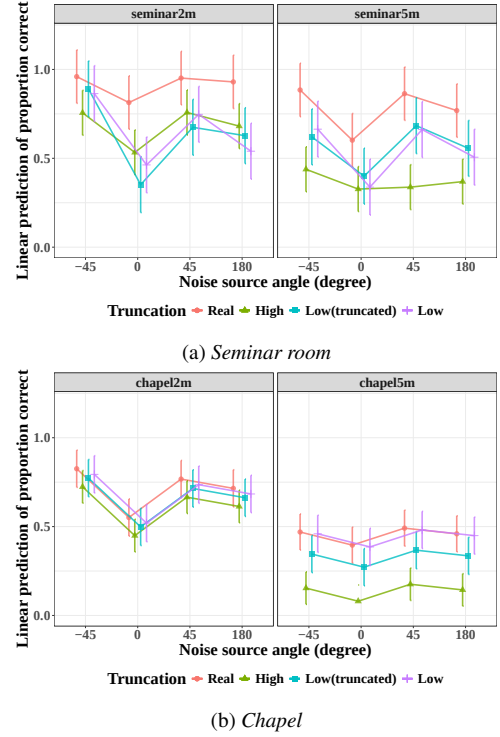


Figure 3: Linear prediction of proportion correct in terms of speech-noise separation split by sound source distance. Error bars show 99% confidence intervals.

3. Results and discussion

For the seminar room, a significant three-way interaction was found between the source distance, noise source angle, and environment ($\chi^2(31) = 293.2$, $p < .0001$). For the chapel, significant two-way interactions were found between the source distance and noise source angle ($\chi^2(3) = 13.88$, $p < .01$), and between the source distance and environment ($\chi^2(3) = 26.12$, $p < .0001$).

Figure 3 shows the predicted models of proportion correct in each room with respect to the noise source angle split by source distance. Table 2 shows the pairwise contrast between the proportion correct scores of the real room test and sound reproduction test (top three blocks) as well as between the scores of the sound reproduction test using different RIR (bottom three blocks). A score from the sound reproduction test being not significantly different from that of the real room test suggests that the spatial sound reproduction was able to reproduce the speech perception in the real room. Similarly a non-significant pair between the environments using different RIR means there was no improvement or degradation of reproducibility by the change of RIR. The results show that spatial sound reproduction using Low RIR successfully reproduced the speech perception in the (real) chapel, but not in the (real) seminar room. The slightly higher noise level in the seminar room Low RIR compared to the chapel Low RIR seen in Figure 2 may have caused the difference. Notable improvement of reproducibility using Low RIR instead of High RIR was observed when the distance was 5 m but not 2 m in both rooms. Because the level of direct sound received by the listener is reduced as the source distance increases, the perception at 5 m would be more sensitive to the reverberation effect caused by the noise floor. Further experi-

Table 2: Pairwise contrasts between room impulse responses. *p*-values in boldface indicate significant pairs ($p < .01$).

contrast	angle	Seminar2m			Seminar5m			Chapel2m			Chapel5m		
		estimate	t.ratio	p.value	estimate	t.ratio	p.value	estimate	t.ratio	p.value	estimate	t.ratio	p.value
Real - High	-45	0.20	2.83	0.03	0.45	6.19	<.0001	0.00	-0.05	1	0.35	4.80	<.0001
	0	0.28	3.89	<.001	0.28	3.82	<.001	0.22	2.95	0.02	0.23	3.18	<.01
	45	0.19	2.69	0.04	0.53	7.30	0	0.14	1.92	0.22	0.32	4.42	<.0001
	180	0.25	3.45	<.01	0.40	5.55	<.0001	0.04	0.56	0.94	0.36	4.90	<.0001
Real - Low	-45	0.10	1.19	0.63	0.22	2.75	0.03	-0.06	-0.67	0.91	-0.04	-0.52	0.95
	0	0.35	4.37	<.0001	0.26	3.29	<.01	0.11	1.37	0.52	0.00	0.00	1
	45	0.20	2.54	0.06	0.20	2.51	0.06	0.05	0.55	0.95	-0.04	-0.51	0.96
	180	0.39	4.85	<.0001	0.26	3.27	<.01	0.02	0.23	1	0.12	1.52	0.43
Real - Low(Truncated)	-45	0.07	0.86	0.83	0.26	3.28	<.01	-0.05	-0.56	0.94	0.18	2.20	0.12
	0	0.46	5.75	<.0001	0.20	2.52	0.06	0.17	2.02	0.18	-0.06	-0.74	0.88
	45	0.28	3.45	<.01	0.18	2.24	0.11	0.01	0.08	1	0.06	0.75	0.88
	180	0.30	3.77	<.01	0.21	2.65	0.04	0.08	0.95	0.78	0.32	3.89	<.001
High - Low	-45	-0.11	-1.47	0.46	-0.23	-3.07	0.012	-0.05	-0.69	0.90	-0.39	-5.22	<.0001
	0	0.07	0.96	0.77	-0.01	-0.15	1	-0.11	-1.43	0.48	-0.23	-3.09	0.011
	45	0.01	0.14	1	-0.32	-4.41	<.0001	-0.10	-1.31	0.56	-0.36	-4.85	<.0001
	180	0.14	1.92	0.22	-0.14	-1.87	0.24	-0.02	-0.30	0.99	-0.23	-3.11	0.011
High - Low(Truncated)	-45	-0.13	-1.83	0.26	-0.18	-2.48	0.06	-0.04	-0.57	0.94	-0.17	-2.27	0.11
	0	0.18	2.47	0.07	-0.07	-0.99	0.76	-0.05	-0.70	0.90	-0.29	-3.89	<.001
	45	0.08	1.14	0.67	-0.35	-4.71	<.0001	-0.14	-1.82	0.26	-0.26	-3.48	<.01
	180	0.05	0.73	0.88	-0.19	-2.54	0.05	0.04	0.49	0.96	-0.04	-0.55	0.95
Low (Truncated) - Low	-45	0.03	0.33	0.99	-0.04	-0.54	0.95	-0.01	-0.11	1	-0.22	-2.68	0.04
	0	-0.11	-1.38	0.51	0.06	0.76	0.87	-0.05	-0.66	0.91	0.06	0.73	0.89
	45	-0.07	-0.90	0.80	0.02	0.27	0.99	0.04	0.47	0.97	-0.10	-1.24	0.60
	180	0.09	1.08	0.70	0.05	0.62	0.93	-0.06	-0.72	0.89	-0.19	-2.33	0.09

Table 3: Pairwise contrasts between noise source angles. *p*-values in boldface indicate significant pairs ($p < .01$). The underlined values are the cases which are likely affected by the ceiling effect.

contrast	implementation	Seminar2m			Seminar5m			Chapel2m			Chapel5m		
		estimate	t.ratio	p.value	estimate	t.ratio	p.value	estimate	t.ratio	p.value	estimate	t.ratio	p.value
0-45	Real	-0.14	-1.79	0.28	-0.26	-3.40	<.01						
	High	-0.22	-3.52	<.01	-0.01	-0.17	1						
	Low	-0.28	-3.52	<.01	-0.32	-4.01	<.001	-0.22	-5.56	<.0001	-0.10	-2.48	0.06
	Low(truncated)	-0.32	-4.00	<.001	-0.28	-3.52	<.01						
(-45)-0	Real	0.15	1.90	0.23	0.28	3.67	<.01						
	High	0.22	3.50	<.01	0.11	1.74	0.30						
	Low	0.40	4.97	<.0001	0.33	4.04	<.001	0.27	7.05	<.0001	0.07	1.92	0.22
	Low(truncated)	0.54	6.68	<.0001	0.22	2.74	0.03						
0-180	Real	-0.12	-1.51	0.43	-0.17	-2.16	0.14						
	High	-0.15	-2.32	0.10	-0.04	-0.66	0.91						
	Low	-0.08	-0.96	0.78	-0.17	-2.09	0.16	-0.16	-4.22	<.001	-0.06	-1.65	0.35
	Low(truncated)	-0.28	-3.41	<.01	-0.16	-1.94	0.21						

ments using RIR generated by numerical simulation controlling the noise floor would be required to confirm these.

One of the key acoustical characteristics of spaces that spatial sound reproduction is able to reproduce is the source angle. A well-known phenomenon in speech perception in noise attributed to source angle is spatial release from masking (SRM) [28], where speech intelligibility improves when the target speech and noise sources are arriving from different angles. The reproducibility of SRM can be measured by looking at the difference of speech intelligibility between the cases where the noise source angle is 0° and other angles. Table 3 shows the results of pair-wise comparison between noise source angles. SRM is observed from the pairs with a significant difference (*p*-values in boldface). Care needs to be taken to interpret the responses at $\pm 45^\circ$, under the seminar2m condition in the real room test (orange lines in the left plot of Figure 3(a) and the values underlined in Table 3) because the responses from these angles very likely suffered from ceiling effect [29] by observing 64 % of participants achieved the proportion correct being one (answered all sentences correctly). By taking this fact into account, it is very likely that SRM was also observed in these cases but not represented by the significance. Overall, in the seminar room, the results suggest the environments with the Low RIR successfully reproduced the SRM observed in the real room. This is not the case with High RIR at 5 m. Previous studies [8, 30] found SRM is reduced under highly reverberant envi-

ronments. The noise floor in the High RIR would have made the virtual sound environment much more reverberant than the real seminar room, resulting in the SRM to be diminished. Truncated RIR also reduced the reproducibility by making the environment less reverberant than the real room. In the chapel, on the other hand, no interaction between the noise source angle and the environment was observed, suggesting spatial sound reproduction replicated the SRM in the real chapel regardless of the RIR used. As the environment was already highly reverberant, the change of reverberation caused by the noise floor would not have had significant effect on speech perception.

4. Conclusion

The current study examined the effect of noise floor in RIR on the reproducibility of speech perception under spatial sound reproduction. The study conducted a listening test both in the real room and under spherical harmonics-based sound reproduction implemented by using RIR with different level of noise floor. The study found using RIR with low noise floor contributed to reproducing speech intelligibility in real room accurately but only when the room was highly reverberant. It also improved the reproducibility regardless of the room when sound sources were located at 5 m from the listener. The effect of SRM was also replicated when RIR with low noise floor was used. Truncating RIR did not show notable benefits to the reproducibility.

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6. References

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